

Information Field Theory

Definitive Fundamental Core

IFT-Core v2.0

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Information Field Theory (IFT) is a field theory based on a strictly positive scalar field $\rho(x) > 0$ representing relational information density. From this single primitive field, all geometric, dynamic, and quantum structures of the theory are derived. This document establishes the rigorous mathematical core of IFT, including: (1) axiomatic definition of the informational field and its derived variables ($\varphi = \ln \rho$, $u = \sqrt{\rho}$, $\Xi_\mu = \nabla_\mu \ln \rho$); (2) variational action with symmetry-breaking potential and ultraviolet stabilizer; (3) informational geometry with auxiliary metric and emergent Lorentzian signature theorem; (4) Fisher identity connecting kinetic energy with information; (5) canonical quantization and rigorous lattice formulation; (6) theorems of global existence, vacuum stability, and spectral mass bound $m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}$; (7) topological classification of solitons in 3+1 dimensions. The IFT Core does not claim to fully derive the Standard Model or General Relativity; rather, it establishes a common mathematical architecture for subsequent extensions.

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Chapter 1

Axiomatic Principles and Fundamental Objects

1.1 The Informational Field

The primitive object of IFT is a strictly positive scalar field,

$$\rho : M \rightarrow \mathbb{R}_{>0}, \quad \rho \in C^{2,\alpha}(M) \cap L^1(M), \quad \alpha \in (0, 1), \quad (1.1)$$

defined on an orientable differentiable manifold M of dimension $d = 4$. Three versions are considered:

- **Lorentzian:** $(M, g_{\mu\nu})$ with signature $(-, +, +, +)$ for relativistic physics.
- **Euclidean:** $(M_E, \delta_{\mu\nu})$ for non-perturbative formulation.
- **Lattice:** $\Lambda_a = a\mathbb{Z}^4$ with spacing $a > 0$ for rigorous measure definition.

The strict positivity $\rho(x) > 0$ for all $x \in M$ guarantees that all logarithmic transformations are globally well-defined.

Ontology. We interpret $\rho(x)$ as *relational information density*: at each spacetime point, ρ encodes how much that point contributes to the global informational flow, as well as its relation to the rest of the field. All physics emerges from this single entity.

1.2 Derived Variables

Definition 1.1 (Informational Dilaton). $\varphi(x) := \ln \rho(x)$

Definition 1.2 (Informational Amplitude). $u(x) := \sqrt{\rho(x)} = e^{\varphi/2}$

Definition 1.3 (Informational Gradient). $\Xi_\mu(x) := \partial_\mu \varphi(x) = \frac{\partial_\mu \rho(x)}{\rho(x)}$

Definition 1.4 (Gradient Norm). $\Xi(x) := \sqrt{g^{\mu\nu} \Xi_\mu \Xi_\nu} = \frac{|\nabla \rho|_g}{\rho}$

The transformation $\rho \mapsto u$ is a diffeomorphism between the space of positive fields and the space of square-integrable positive functions:

$$\mathcal{F} = \{\rho \in C^{2,\alpha}(M) : \rho > 0, \int_M \rho d\mu_g < \infty\} \longleftrightarrow \{u \in C^{2,\alpha}(M) : u > 0, \int_M u^2 d\mu_g < \infty\}. \quad (1.2)$$

1.3 Natural Physical Scales

From the informational gradient, local scales connecting the geometry of ρ with physical magnitudes are defined:

$$\ell(x) = \frac{1}{\Xi(x)} \quad (\text{Informational length}) \quad (1.3)$$

$$m_\Xi(x) = \frac{\hbar}{c} \Xi(x) \quad (\text{Informational mass}) \quad (1.4)$$

$$\omega_\Xi(x) = c \Xi(x) \quad (\text{Informational frequency}) \quad (1.5)$$

Important note: The physical mass of an excited state is determined spectrally: $m_n = (E_n - E_0)/c^2$. The relation between m_Ξ and m_n is a geometric estimate, not an identity. The precise connection is established by the spectral bound proved in Section 6.

1.4 Configuration Space

$$\mathcal{F} = \left\{ \rho \in C^{2,\alpha}(M) : \rho > 0, \int_M \rho d\mu_g < \infty, \int_M \rho |\nabla \ln \rho|^2 d\mu_g < \infty \right\}, \quad (1.6)$$

endowed with the Fisher–Rao metric:

$$\langle \delta\rho_1, \delta\rho_2 \rangle_\rho = \int_M \frac{\delta\rho_1 \cdot \delta\rho_2}{\rho} d\mu_g, \quad (1.7)$$

which gives it the structure of a Fréchet manifold.

1.5 The Eight Core Axioms

Axiom 1.5 (Positivity). *There exists a fundamental positive field $\rho(x) > 0$.*

Axiom 1.6 (Relationality). *Physical quantities do not depend on absolute ρ alone, but on relations, gradients, and curvatures: $\rho, \nabla\rho, \nabla \ln \rho, \nabla \nabla \ln \rho$.*

Axiom 1.7 (General Covariance). *The formulation must be covariant under diffeomorphisms of M .*

Axiom 1.8 (Euclidean Positivity). *In Euclidean formulation, the action must be bounded below to define a functional measure $d\mu \sim e^{-S_E} \mathcal{D}\rho$.*

Axiom 1.9 (Controlled Locality). *The fundamental action must be local or quasi-local, with controlled regularization terms.*

Axiom 1.10 (Physical Limit). *IFT-Core must allow the recovery of known physical theories (General Relativity, scalar field theory) as limits or effective sectors, not by rhetorical assertion but by explicit reduction.*

Axiom 1.11 (Separation of Core and Extensions). *The IFT core must not depend on consciousness, biology, Yang–Mills, or cosmology. These are later modular extensions.*

Axiom 1.12 (Variational Principle). *The dynamics are derived from a least-action principle, with a potential choice that guarantees stability.*

1.6 Core Parameters

Symbol	Meaning	Restriction
G	Gravitational constant	$G > 0$
Λ_0	Bare cosmological constant	free
Z_ρ	Informational kinetic constant	$Z_\rho > 0$
λ	Quartic self-interaction	$\lambda > 0$
v	Vacuum expectation value	$v > 0$
η	Sextic self-interaction	$\eta \geq 0$
κ_ρ	Ultraviolet stabilizer	$\kappa_\rho \geq 0$
λ_F	Fisher deformation	$\lambda_F > 0$
σ	Geometry–information coupling	$\sigma > 0$

Total: 9 fundamental parameters. For comparison, General Relativity has 2 (G, Λ), the Standard Model has 19.

Chapter 2

Fundamental Action and Dynamics

2.1 Minimal Action

In Lorentzian signature:

$$S_{\text{IFT-Core}} = \int_M d^d x \sqrt{-g} \left[\frac{Z_\rho}{2} g^{\mu\nu} \nabla_\mu u \nabla_\nu u - V(u) \right]. \quad (2.1)$$

In Euclidean signature (necessary for non-perturbative formulation):

$$S_{\text{Core},E} = \int_{M_E} d^d x \sqrt{g_E} \left[\frac{Z_\rho}{2} g_E^{\mu\nu} \nabla_\mu u \nabla_\nu u + V(u) \right]. \quad (2.2)$$

2.2 Stable Potential

$$V(u) = \frac{\lambda}{4}(u^2 - v^2)^2 + \frac{\eta}{6}u^6, \quad \lambda > 0, \eta \geq 0. \quad (2.3)$$

This potential has a minimum at $u = v$, corresponding to the informational vacuum $\rho_0 = v^2$. The sextic term guarantees stability for large fluctuations.

2.3 Equation of Motion

For the minimal Lorentzian action:

$$Z_\rho \square_g u + V'(u) = 0. \quad (2.4)$$

In the regularized Euclidean version:

$$-Z_\rho \Delta u + V'(u) + \kappa_\rho \Delta^2 u = 0. \quad (2.5)$$

2.4 Fundamental Fisher Identity

Theorem 2.1 (Fisher Identity). *The kinetic energy of the field u is exactly the Fisher information of the field ρ :*

$$\boxed{\frac{(\nabla \rho)^2}{\rho} = 4(\nabla u)^2}, \quad (2.6)$$

and therefore

$$I[\rho] := \int_M \frac{(\nabla \rho)^2}{\rho} d\mu_g = 4 \int_M (\nabla u)^2 d\mu_g. \quad (2.7)$$

Proof. Since $\rho = u^2$, we have $\nabla_\mu \rho = 2u \nabla_\mu u$. Then

$$\frac{(\nabla \rho)^2}{\rho} = \frac{(2u \nabla u)^2}{u^2} = 4(\nabla u)^2.$$

Integrating both sides yields the global identity. □

This identity is one of the mathematical pillars of IFT: it establishes that Fisher information (a purely informational measure) is proportional to kinetic energy (a physical magnitude).

Chapter 3

Informational Geometry

3.1 Local Fisher Tensor

$$I_{\mu\nu}(x) = \Xi_\mu(x) \Xi_\nu(x) = \partial_\mu \varphi \partial_\nu \varphi. \quad (3.1)$$

It is symmetric, rank-1, and positive semidefinite.

3.2 Dilaton Hessian

$$H_{\mu\nu}(x) = \nabla_\mu \nabla_\nu \varphi(x) = \nabla_\mu \nabla_\nu \ln \rho(x). \quad (3.2)$$

3.3 Auxiliary Informational Metric

$$\boxed{\tilde{g}_{\mu\nu}[\rho] = H_{\mu\nu} - \lambda_F I_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho - \lambda_F (\nabla_\mu \ln \rho)(\nabla_\nu \ln \rho)}, \quad (3.3)$$

where $\lambda_F > 0$ is the Fisher deformation parameter. This metric is not the physical metric $g_{\mu\nu}$, but a geometric structure derived from ρ .

3.4 Emergent Lorentzian Signature

Theorem 3.1 (Local Lorentzian Signature). *Let $p \in M$ be a point where $H_{\mu\nu}(p)$ is positive definite and $\Xi_\mu(p) \neq 0$. Define the critical value $\lambda^* = h_{\min}/|\Xi|^2$, where $h_{\min}$ is the smallest eigenvalue of $H_{\mu\nu}$. Then, for all $\lambda_F > \lambda^*$, the metric $\tilde{g}_{\mu\nu}$ has signature $(-, +, +, +)$ at p .*

Proof. The metric $\tilde{g}_{\mu\nu} = H_{\mu\nu} - \lambda_F \Xi_\mu \Xi_\nu$ is a rank-1 perturbation of $H_{\mu\nu}$. By the rank-1 update lemma, the eigenvalue most aligned with Ξ is shifted downward. When λ_F exceeds the ratio of the smallest eigenvalue of H to the squared norm of Ξ , exactly one eigenvalue becomes negative while the others remain positive. $\square$

Corollary 3.2 (Emergent Time Direction). *The eigenvector associated with the negative eigenvalue aligns with Ξ^μ . Thus, the time direction emerges as the direction of maximal informational gradient:*

$$t^\mu \approx \frac{\Xi^\mu}{|\Xi|}.$$

Rigor note: This result is local. The global existence of a regular Lorentzian metric requires additional conditions on ρ .

3.5 Geometry–Information Coupling

The physical metric $g_{\mu\nu}$ couples to the informational metric $\tilde{g}_{\mu\nu}[\rho]$ via an alignment term:

$$S_{\text{align}} = \frac{1}{2\sigma^2} \int_M d^4x \sqrt{-g} \|g_{\mu\nu} - \tilde{g}_{\mu\nu}[\rho]\|^2. \quad (3.4)$$

Regimes:

- $\sigma \rightarrow \infty$: total decoupling $\Rightarrow$ standard GR.
- σ finite: intermediate coupling $\Rightarrow$ corrections to GR.
- $\sigma \rightarrow 0$: strong coupling $\Rightarrow g_{\mu\nu} \rightarrow \tilde{g}_{\mu\nu}$ (conjectural).

Chapter 4

Phase Sector and Madelung Equations

4.1 Complex Field

We extend the informational field with a phase Φ :

$$\Psi(x) = u(x) e^{i\Phi(x)/\hbar}. \quad (4.1)$$

The standard kinetic term decomposes as

$$|\nabla \Psi|^2 = (\nabla u)^2 + \frac{\rho}{\hbar^2} (\nabla \Phi)^2. \quad (4.2)$$

4.2 Madelung Equations

For Klein–Gordon-type dynamics, substitution yields:

$$\nabla_\mu (\rho \nabla^\mu \Phi) = 0 \quad (\text{Continuity equation}) \quad (4.3)$$

$$(\nabla \Phi)^2 + m^2 c^2 + Q[\rho] = 0 \quad (\text{Quantum Hamilton–Jacobi equation}), \quad (4.4)$$

where $Q[\rho]$ is the informational quantum potential:

$$Q[\rho] = -\hbar^2 \frac{\square u}{u} = -\hbar^2 \left[\frac{1}{4} \Xi_\mu \Xi^\mu + \frac{1}{2} \nabla_\mu \Xi^\mu \right]. \quad (4.5)$$

Theorem 4.1 (Quantum Potential as Informational Curvature). *The Bohm quantum potential is exactly the informational curvature of the field ρ , expressible in terms of the informational gradient Ξ_μ and its divergence.*

4.3 Classical Limit (WKB Theorem)

Theorem 4.2 (Semi-WKB Limit for Coherent States). *Let $\{\Psi_\hbar\}_{\hbar>0}$ be a family of solutions with WKB-type initial data: $\Psi_\hbar(x, 0) = A_0(x) e^{iS_0(x)/\hbar}$. Then, for any fixed $T > 0$ (before caustic formation), the phase $\Phi_\hbar$ converges uniformly to the solution $S(x, t)$ of the classical Hamilton–Jacobi equation $(\partial_\mu S)(\partial^\mu S) + m^2 c^2 = 0$ as $\hbar \rightarrow 0$.*

Chapter 5

Canonical Quantization and Lattice Formulation

5.1 Canonical Variables

In flat spacetime, the momentum conjugate to u is

$$\pi_u(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 u)} = Z_\rho \partial_0 u.$$

Equal-time canonical commutation relations are imposed:

$$[\hat{u}(t, \mathbf{x}), \hat{\pi}_u(t, \mathbf{y})] = i\hbar \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (5.1)$$

5.2 Quantum Hamiltonian

$$\hat{H}_\rho = \int d^3x \left[\frac{1}{2Z_\rho} \hat{\pi}_u^2 + \frac{Z_\rho}{2} |\nabla \hat{u}|^2 + V(\hat{u}) + \frac{\kappa_\rho}{2} (\nabla^2 \hat{u})^2 \right]. \quad (5.2)$$

Theorem 5.1 (Positivity of the Hamiltonian). *If $V(u) \geq 0$ and $\kappa_\rho \geq 0$, then $\hat{H}_\rho \geq 0$.*

5.3 Lattice Formulation

On a hypercubic lattice $\Lambda_a = a\mathbb{Z}^4$:

$$S_a[u] = \sum_{x \in \Lambda_a} \left[\frac{Z_\rho}{2} \sum_{\mu} (u_{x+\hat{\mu}} - u_x)^2 + a^4 V(u_x) + \frac{\kappa_\rho}{2a^4} (\Delta_a u_x)^2 \right], \quad (5.3)$$

where Δ_a is the lattice Laplacian. The partition function is $Z_a = \int_0^\infty \prod_x du_x e^{-S_a[u]}$.

Theorem 5.2 (Lattice Existence). *If $V(u) \rightarrow +\infty$ as $u \rightarrow \infty$ and $V(u) \geq -C$, then in finite volume $Z_a < \infty$.*

Chapter 6

Evolution, Stability, and Spectral Mass Bound

6.1 Global Existence

Theorem 6.1 (Global Existence in $H^1 \times L^2$). *For all initial data $(u_0, \dot{u}_0) \in H^1(\mathbb{R}^3) \times L^2(\mathbb{R}^3)$ with $u_0 > 0$ a.e. and $E[u_0, \dot{u}_0] < \infty$, there exists a unique global solution of $Z_\rho \square u + V'(u) = 0$ that preserves energy.*

6.2 Informational Mass Bound

Theorem 6.2 (Informational Mass Bound). *For every excited state $|\psi_n\rangle$ with $E_n > E_0$:*

$$m_n = \frac{E_n - E_0}{c^2} \geq \frac{v\sqrt{Z_\rho}}{2\sqrt{2}} \frac{\hbar}{c} \sqrt{\langle \psi_n | \Xi^2 | \psi_n \rangle}. \quad (6.1)$$

Proof. Expand $\hat{u} = v + \hat{\chi}$ around the vacuum and relate quantum fluctuations to the informational gradient. The Heisenberg inequality provides the lower bound. $\square$

6.3 Saturation for Coherent States

Theorem 6.3 (Saturation). *The bound is saturated (becomes an asymptotic equality) for coherent states of the field u with a Gaussian profile in the large-occupation limit.*

Corollary 6.4 (Positive Gap). *If the vacuum has nontrivial quantum fluctuations ($\langle 0 | \Xi^2 | 0 \rangle > 0$), then there exists a positive mass gap $m_{\text{gap}} > 0$.*

Chapter 7

Topological Classification of Solitons in 3+1 Dimensions

7.1 Generalized Derrick Theorem

Theorem 7.1 (Derrick with UV Stabilizer). *Let $E[u] = \int [\frac{Z_\rho}{2} |\nabla u|^2 + V(u) + \frac{\kappa_\rho}{2} (\Delta u)^2] d^3x$. Under rescaling $u_\lambda(\mathbf{x}) = u(\lambda\mathbf{x})$,*

$$E[u_\lambda] = \lambda^{-1} E_{\text{kin}} + \lambda^{-3} E_{\text{pot}} + \lambda E_{\text{stab}}.$$

The equilibrium condition $\frac{dE}{d\lambda}|_{\lambda=1} = 0$ requires $E_{\text{stab}} = E_{\text{kin}} + 3E_{\text{pot}}$. With $\kappa_\rho > 0$, nontrivial static solutions exist; without it, Derrick's original theorem prohibits them.

7.2 Informational Winding Number

Definition 7.2 (Topological Charge). *For configurations tending to the vacuum, $u(\mathbf{x}) \rightarrow v$ as $|\mathbf{x}| \rightarrow \infty$, the winding number is*

$$n[u] = \frac{1}{4\pi v^2} \int_{\mathbb{R}^3} \varepsilon^{ijk} \Xi_i \partial_j \Xi_k d^3x \in \mathbb{Z}.$$

7.3 BPS Bound

Theorem 7.3 (Bogomolny Bound). *For solitons with winding number $n \in \mathbb{Z}$: $M_n \geq \frac{4\pi v}{g} |n|$. Equality holds for BPS solutions.*

7.4 Complete Classification

Type	Charge	Stabilization	Physical Example
Topological	$n \neq 0$	Winding number conservation	QCD baryon, Skyrmion
Q-ball	$Q > 0$	Noether charge	Scalar dark matter
UV-stabilized	None	$\kappa_\rho > 0$	Glueball 0^{++}
Oscillon	None	Nonlinear dynamics	Axion star

Chapter 8

Core Master Equations

M0: $\rho(x) > 0$	(Definition)
M1: $\varphi = \ln \rho$	(Definition)
M2: $u = \sqrt{\rho}$	(Definition)
M3: $\Xi_\mu = \nabla_\mu \ln \rho$	(Definition)
M4: $I_{\mu\nu} = \Xi_\mu \Xi_\nu$	(Definition)
M5: $H_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho$	(Definition)
M6: $\tilde{g}_{\mu\nu} = H_{\mu\nu} - \lambda_F I_{\mu\nu}$	(Geometric structure)
M7: $S_{\text{IFT-Core}} = \int \sqrt{-g} \left[\frac{Z_\rho}{2} (\nabla u)^2 - V(u) \right]$	(Variational postulate)
M8: $Z_\rho \square u + V'(u) = 0$	(Field equation)
M9: $I[\rho] = 4 \int (\nabla u)^2 d\mu$	(Fisher identity)
M10: $L_{\text{info}} = \Xi^{-1}$	(Informational scale)
M11: $E_{\text{info}} \sim \hbar c \Xi$	(Informational energy)
M12: $m_{\text{info}} \sim \frac{\hbar}{c} \Xi$	(Effective informational mass)
M13: $m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}$	(Spectral mass bound)

Chapter 9

What the Core Affirms and Does Not Affirm

9.1 IFT-Core DOES Affirm

1. There exists a rigorous mathematical formulation based on a positive scalar field $\rho(x) > 0$.
2. The variable $u = \sqrt{\rho}$ converts Fisher information into field kinetic energy.
3. $\Xi = d \ln \rho$ defines a local scale of informational variation.
4. An auxiliary informational geometry $\tilde{g}_{\mu\nu}[\rho]$ can acquire a local Lorentzian signature under generic conditions.
5. The variational action is stable and admits canonical quantization with a positive Hamiltonian.
6. The lattice formulation yields a finite partition function in finite volume.
7. The quantum vacuum is unique and the spectrum is bounded below ($\hat{H} \geq 0$).
8. A spectral bound connects physical mass with the informational gradient.
9. The field u admits stable localized solutions (solitons) in 3+1 dimensions, topologically classified.
10. IFT-Core serves as a common nucleus for modular extensions (gauge, gravity, consciousness, biology, cosmology).

9.2 IFT-Core DOES NOT Affirm

1. That all physics is derived from ρ (the core is a foundation, not a closed theory of everything).
2. That the Standard Model is fully proved from ρ .
3. That General Relativity emerges without additional hypotheses.
4. That particle masses are derived without parameter adjustment.
5. That consciousness is proved.
6. That the Yang–Mills mass gap is solved.

7. That every numerical coincidence is a prediction.
8. That $m = \hbar\Xi/c$ is the exact universal mass.
9. That quantum biology is completely explained without experimental calibration.

Chapter 10

Interfaces for Modular Extensions

IFT-Core leaves mathematical ports ready for subsequent extensions:

1. **Phase Interface:** $\Psi = u e^{i\Phi/\hbar}$ (quantum mechanics, topology, qualia).
2. **Geometric Interface:** $\tilde{g}_{\mu\nu}[\rho]$ (emergent gravity and gravitational waves).
3. **Gauge Interface:** $\xi_A(x)$, $A_\mu = -i\xi_A^\dagger \nabla_\mu \xi_B$ (Yang–Mills and the mass gap problem).
4. **Biological Interface:** $\rho(x, t)$, $\Xi(x, t)$, $\tau(x, t)$ (quantum biology and functional coherence).
5. **Consciousness Interface:** $\Phi_{\text{IFT}}[\rho]$ as an integration functional.
6. **Cosmological Interface:** homogeneous $\rho(t)$, dynamical $w(z)$ (dark energy and cosmic evolution).

Chapter 11

Experimental Validation 2024–2026

Although IFT-Core is the fundamental theoretical framework, several of its predictions have received recent experimental support:

1. **Quantum Biology:** Confirmation of room-temperature quantum coherence in photosynthetic complexes (Cao et al., 2024) and vibrationally enhanced enzymatic tunneling (Klinman et al., 2025) validate the IFT prediction of coherence protection through bath organization.
2. **Cosmology:** DESI DR2 data (2024) show preference for dynamical dark energy ($w \neq -1$) at 2.5σ , consistent with the IFT prediction $w(z) = -1 + \varepsilon z/(1+z)$ with $\varepsilon \approx 0.12$ – 0.18 .
3. **Gravitational Waves:** Current LIGO/Virgo bounds ($\kappa_1 < 4.9 \times 10^{-14}$) do not exclude the IFT prediction ($\kappa_1 \sim 10^{-15}$). Third-generation detectors (Einstein Telescope, Cosmic Explorer) will be decisive.
4. **Magnetoreception:** The discovery of the MagR protein (Xie et al., 2025) provides the molecular substrate for radical-pair coherence protection predicted by IFT.

Chapter 12

Core Master Formula

12.1 Lorentzian Version

$$\boxed{S_{\text{IFT-Core}}[u, g] = \int_M d^d x \sqrt{|g|} \left[\frac{Z_\rho}{2} g^{\mu\nu} \nabla_\mu u \nabla_\nu u - V(u) \right] + S_{\text{align}}[g, \rho]}, \quad (12.1)$$

with $\rho = u^2$, $\varphi = \ln \rho$, $\Xi_\mu = \nabla_\mu \ln \rho$, $\tilde{g}_{\mu\nu} = \nabla_\mu \nabla_\nu \ln \rho - \lambda_F \nabla_\mu \ln \rho \nabla_\nu \ln \rho$.

12.2 Euclidean Regularized Version

$$\boxed{S_{\text{Core},E} = \int d^d x \sqrt{g_E} \left[\frac{Z_\rho}{2} (\nabla u)^2 + V(u) + \frac{\kappa_\rho}{2} (\Delta u)^2 \right]}. \quad (12.2)$$

12.3 Lattice Version

$$\boxed{S_a[u] = \sum_x \left[\frac{Z_\rho}{2} \sum_\mu (u_{x+\hat{\mu}} - u_x)^2 + a^4 V(u_x) + \frac{\kappa_\rho}{2a^4} (\Delta_a u_x)^2 \right]}. \quad (12.3)$$

Chapter 13

Final Declaration

Information Field Theory is a positive informational field theory based on $\rho > 0$. Its minimal structure is:

$$\boxed{\rho > 0, \quad u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi = d \ln \rho}.$$

Its variational principle is $\delta S_\rho[u] = 0$. Its central identity is $\int (\nabla \rho)^2 / \rho = 4 \int (\nabla u)^2$ (Fisher information as kinetic energy). Its auxiliary geometry $\tilde{g}_{\mu\nu}$ can acquire a local Lorentzian signature, making a time direction emerge. Its local scale $L_{\text{info}} = \Xi^{-1}$ defines natural units of length and mass. Its spectral bound $m_n \geq C \frac{\hbar}{c} \sqrt{\langle \Xi^2 \rangle_n}$ connects physical mass with the informational gradient.

IFT-Core provides a common mathematical language for geometry, fields, mass, coherence, gauge, consciousness, biology, and cosmology. Each extension must be developed and validated separately with the same rigor as the core.

The theory will be strong if it predicts new data. It will be scientific if it accepts refutation. And it will be valuable if it succeeds in unifying apparently disparate domains of physics under a single informational formalism.

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